E.67 standalone proof review

Statement E.67 (Proved). The original Mordell sign and the complete finite Gaussian norm comparison. *Retain the separate original orbifold actions Ω_3, Ω_4 and their Ferrari homology matrices B_3, B_4 from E.64.28–29. For every multiplier-one normalized gentle pair, the action B_3 on the positive E.1 line $\mathbf{F}_3(1,-1)$ is the identity. Consequently, for the full-GT comparison characters of E.64,*

$$\epsilon_3(C_3) = \epsilon_3(D_3), \qquad \delta = \epsilon_3(D_3) + \epsilon_2(D_4) + \epsilon_2(B_4). \tag{E.67.1}$$

The latter is an identity of the specified characters; it does not assign the remaining sum on full GT.

The Gaussian term has a finite evaluation formula with its actual word and origin retained. Put $V = \mathbf{F}_3^2$ and

$$R = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \qquad S = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{E.67.2}$$

Use $(v,A)(w,B) = (v + Aw, AB)$ in the following two finite groups:

	group	x	y	z
M	$V \rtimes \langle R \rangle$	$(0,R)$	(e_1, R^3)	(e_1, R^2)
G	$V \rtimes \langle S \rangle$	(e_1, S)	$(-e_2, S^2)$	$(0,S)$.

(E.67.3)

These are the actual respective quotients of the two triangles by $[K,K]K^3$, with orders 54 and 36. If v is the translation component of $f(x,y)$ in G*, then*

$B_4 \bmod 3$	v	$\epsilon_2(B_4)$
I	$(0,0)$	0
$-I$	$(2,0)$	0
S	$(1,1)$	1
$-S$	$(1,2)$	1

$$B_4 e_1 = e_1 - v. \tag{E.67.4}$$

In particular the bit is $v_2^2 \in \{0,1\} \subset \mathbf{F}_3$, read in $\mathbf{F}_2$.

There is also a complete finite correction-image comparison over the actual E.1 source. Let $N = N_B$, $K_6 = \overline{N'N^6}$, and evaluate the same free word in the following ordered pairs, first in M *and then in* G*:*

$$(x,y),(y,x),(z,x),(x,z),(y,z),(z,y). \tag{E.67.5}$$

Write Θ for this twelve-evaluation map. On K_6 its values are translations. With $w = (1,-1)$ its exact image is

$$\begin{aligned} \Theta(K_6) = \Theta(N') &= \mathcal{M} \times \mathcal{G}, \\ \mathcal{M} &= \{(aw, aw, bw, cw, -cw, -bw) : a,b,c \in \mathbf{F}_3\}, \\ \mathcal{G} &= \{(u,v,t,-t,-Sv,-Su) : u,v,t \in V\}. \end{aligned} \tag{E.67.6}$$

Thus the common image has order 3^9, and its two projection ranks are 3 and 6. Every vector in this displayed product has an actual word preimage in $N' \subset \widehat{F}'$.

In each of the 24 original E.1 word fibres, and separately in each of the 24 E.17 word fibres used in E.62 and E.66, there are exactly four attainable twelve-evaluation vectors which satisfy all the two-cycle and three-cycle equations and give determinant-one automorphisms of both groups in (E.67.3)*. They have one uniquely specified framed Mordell target and all*

four Gaussian targets in (E.67.4)*. The two origin lists are evaluated separately; no equality of their G_6 values is used.*

These are complete correction-image fibres. For a retained origin f_j and one of its four values t, choose the saved literal correction $c_{j,t} \in N'$. The entire corresponding word fibre in $f_j(K_6 \cap \widehat{F}')$ is exactly

$$f_j c_{j,t}(K_6 \cap \widehat{F}' \cap \ker \Theta). \tag{E.67.7}$$

The corrected pair induces an automorphism of the actual two-factor image $\mathrm{M} \times \mathrm{G}$ *of order* 1944*, so its map onto that image is generating. These assertions retain their specified finite source and equations; they do not assert a full-GT lift or an automorphism of an unevaluated larger joint source.*

Proof. We prove the two quotient maps and their frame equations before identifying the complete common correction image.

The actual Mordell quotient and its positive sign. The original triangle in E.64 has $xyz = 1$, $y^2 = z^3 = 1$. Its free kernel K has the Ferrari generators

$$k_1 = zx^{-2}, \qquad k_2 = x^4 z. \tag{E.67.8}$$

The cusp identity is $[k_1, k_2]x^6 = 1$. Hence $x^6 = 1$ after abelianizing K, and the deck action splits there. Substitution in (E.67.3) sends k_1, k_2 to e_1, e_2, respectively. It sends the deck generator to R, of order six. The resulting epimorphism has exactly the kernel $[K, K]K^3$: the induced map on K is its abelianization modulo three and the deck quotient is unchanged. This proves the stated order-54 comparison.

At multiplier one, E.64.28 fixes x and sends z to $f(z, x)zf(z, x)^{-1}$. The conjugator has trivial deck coordinate because $f \in \widehat{F}'$. Write its translation as $a \in V$. The induced matrix B_3 commutes with R, since x is fixed. Comparing the two images of z gives the exact equality

$$(B_3 - I)e_1 = (I - R^2)a. \tag{E.67.9}$$

The image of $I - R^2$ is $\mathbf{F}_3 w$, where $w = (1, -1)$, and e_1 has nonzero image in $V/\mathbf{F}_3 w$. Direct solution of $BR = RB$ gives $B = AI + B_0 R$. Its scalar both on $\mathbf{F}_3 w$ and on the indicated one-dimensional quotient is $A - B_0$. Equality (E.67.9) makes that scalar one. This proves the positive sign with the actual deck-zero frame retained. The determinant-one condition from the primitive boundary commutator in E.64.29 leaves the three matrices in this positive centralizer coset.

The original Tate action used here is the separate map in [1, Proposition 4.4]; the comparison with C_3 is the actual quotient $D_3 = C_3 B_3^{-1}$ of E.64. Thus its sign gives the first identity in (E.67.1) without identifying the two actions. Applying the same identity $C_2 = D_4 B_4$ and then the exact parity definition in E.65 gives the second.

The actual Gaussian quotient and all four finite actions. In the quartic subgroup of E.64 write $z = d$. Its deck generator acts by S, and the retained Ferrari expressions give

$$x = \xi_1 d^{-3}, \qquad y = \xi_2^{-1} d^2, \qquad d^4 = 1 \tag{E.67.10}$$

after taking the stated centre quotient. Abelianizing $K = \langle \xi_1, \xi_2 \rangle$ modulo three therefore gives exactly the Gaussian row of (E.67.3). In that quotient $x^2 y = e_1$ and $z^2 y = e_2$; these also prove surjectivity and the exact kernel comparison.

For an actual multiplier-one orbifold action, B_4 commutes with S and has determinant one by E.64.29. Solving these two equations over $\mathbf{F}_3$ gives exactly $I, -I, S, -S$. Put $f(x, y) =$

(v, I). Its two-cycle partner is $f(y, x) = (-v, I)$, so the action sends y to $(-v, I)y(v, I)$. Evaluating $x^2y = e_1$, with x fixed, gives $B_4e_1 = e_1 - v$. This proves the complete table (E.67.4), including the parity and its sign.

All four finite automorphisms have explicit frames. Given $B \in \{I, -I, S, -S\}$, fix x and put

$$z_B = ((I - B)e_1, S), \qquad y_B = (z_B x)^{-1}. \tag{E.67.11}$$

The resulting map acts by B on V. Its multiplication law is preserved, and its inverse uses B^{-1}, so it is an automorphism of the full order-36 group. Equivalently, the certificate checks its entire permutation on those 36 elements. The matrices $I - S$ and $I - S^2$ are invertible, so the conjugating translations h_z, h_y with $z_B = h_z z h_z^{-1}$ and $y_B = h_y y h_y^{-1}$ are unique.

In the Mordell group, a positive determinant-one matrix B instead gives $y_B = (Be_1, R^3)$ and $z_B = (Be_1, R^2)$. The y frame is unique because $I - R^3$ is invertible. The z frames form a coset of $\ker(I - R^2)$, of order three. Consequently there are exactly nine framed Mordell targets, over its three possible matrices. For either group the complete six translation values dictated by these frames and the two-cycle and three-cycle equations are

$$(-h_y,\ h_y,\ h_z,\ -h_z,\ h_y - h_z,\ h_z - h_y). \tag{E.67.12}$$

This supplies all $9 \cdot 4$ candidate pairs before intersecting with any old-source fibre; the z-frame ambiguity in the Mordell factor is fully retained.

The complete actual correction image. Use the 97 literal free generators n_i of N_B from the previously certified E.1 source, with the unchanged word list in the present input. In every Mordell substitution their images have even deck coordinate, so lie in $V \rtimes \langle R^2 \rangle$. This group has exponent three: $I + R^2 + R^4 = 0$. In every Gaussian substitution their images lie in $V \rtimes \langle S^2 \rangle = V \rtimes \langle -I \rangle$, whose exponent divides six. Thus every sixth power from N_B is killed in all twelve maps. This proves $\Theta(K_6) = \Theta(N'_B)$, including the closed profinite subgroups.

Every commutator image is a translation. The complete calculation evaluates all $\binom{97}{2} = 4656$ literal commutators, retains a word preimage for each pivot, and closes their span under each of the 97 actual conjugation matrices. The initial span already has dimension nine and is stable under all these conjugations. It is therefore the entire image of N'_B, since those commutators normally generate N'_B. The saved nine pivot words are each replayed in all twelve maps.

The two projection ranks are three and six. More explicitly, row reduction gives exactly $\mathcal{M}$ and $\mathcal{G}$ in (E.67.6). The separate small linear certificate checks both inclusions between the nine saved rows and the displayed parametrization. Since the common rank is the sum of the projection ranks, the image is their full product on this actual common domain. In particular every Gaussian correction can be made while preserving all six Mordell values.

There is an additional exact upper-bound check on the Gaussian side. Its displayed six-dimensional space contains the value of $[x, y]$ and is stable under the two actual free-generator conjugations in the six substitutions. It therefore contains the image of the whole $\widehat{F}'$. The already computed image of N'_B is the entire space, proving equality. This uses the literal free-source maps, rather than an identification with one of the cyclic distribution maps used elsewhere.

Complete finite fibres and generating maps. The homogeneous two-cycle and three-cycle equations have zero intersection with $\mathcal{M}$. Indeed in (E.67.6) the first two-cycle equation gives $a = 0$, the second gives $c = -b$, and the three-cycle equation gives $b - c = 0$. Hence $a = b = c = 0$. Thus any given origin has at most one framed Mordell norm target. Testing the complete nine-target list against the actual affine origin coset gives one in each of the 24 E.1 cases and, independently, in each of the 24 E.17 cases.

For the Gaussian space the homogeneous norm equations give $v = -u$ and $t = -(I + S)u$. Intersecting with the determinant-one automorphism condition gives exactly the four values of $u = f(x, y)$ in (E.67.4). Each of the actual origin vectors lies in $\mathcal{G}$, so all four absolute targets are reached. Together with the full-product equality this proves the four-point correction-image fibre. The certificate also constructs and replays one literal N'_B correction for every one of the resulting 96 values in each origin list. Applying the homomorphism Θ to two such words proves the exact coset formula (E.67.7).

The literal pair $(x_{\mathrm{M}}, x_{\mathrm{G}})$, $(y_{\mathrm{M}}, y_{\mathrm{G}})$ generates the full product M × G: enumeration of its actual subgroup gives $54 \cdot 36 = 1944$ elements. For every retained corrected word, the new two-factor pair is the image of this pair under the product of the two constructed automorphisms. Thus the corrected source map is also onto this specified product.

Preserved attempts and reusable maps. Some original E.1 representatives fail a Mordell three-cycle equation before correction; their indices and values are retained. Other fixed representatives fail the Gaussian action or its determinant-one condition. The proof above uses the complete affine correction image, so none of those fixed-word failures is used to exclude its full fibre.

The authoritative programs share the prefix `bc_orbifold_CM_`: their suffixes are `characters.py`, `six.py`, `linear.py`, and `export.py`. The export retains the two literal free-generator permutations on 108 points, all 97 original N_B permutations, the nine correction pivots, the complete framed targets, and both distinct origin lists. Its right-action convention is $p(gh) = p(h)[p(g)]$; the affine action is $t \mapsto A^{-1}(t - v)$. Each literal source word is replayed in that convention. This gives exact inputs for further common-source intersections, without assigning a common quotient with another three-primary cover. □

References

[1] Hiroaki Nakamura, Hiroshi Tsunogai, and Seidai Yasuda. Harmonic and equianharmonic equations in the grothendieck–teichmueller group. iii. *Journal of the Institute of Mathematics of Jussieu*, 9(2):431–448, 2010. verified: opened the complete author-uploaded primary PDF; Proposition 2.19, Proposition 4.4, Proposition 4.13, Theorem 4.16, and the scope statement on printed p.432. Proposition 4.4 is stated for GT; Theorem 4.16 is stated for Galois elements.